

# Model re-entry — operations order (OPORD)

Last synced: 2026-07-27  
Owner: Charles + Cursor agent (not a SCOUT handoff)  
Status: Executed — archive, 540 surface, Pass B exports on disk.

## Intent

Re-entry sim work mirrors the document re-entry pattern: **park old lab notebooks**, build a **thin 540\_\* surface**, reuse **tier1\_\*** engines. Revolution = new problem approach and entry point; evolution = import battle-tested libraries.

Charles authored the three-step pipeline: assign → **S<sub>i</sub>** → top **K**. Alex endorsed it.

User-facing assignment knob: **ρ (rho)** = assortativity (0 = max mixing; ρ↑ = sharper match). Legacy **τ (temperature)** in archived docs = inverse mixing; **ρ = 1, σ = 0.65** maps to old **τ ≈ 0.65**.

## Phase 0 — Preconditions

| #   | Step                                          | Owner   |
|-----|-----------------------------------------------|---------|
| 0.1 | Proceed in this chat (Charles + Cursor agent) | Charles |
| 0.2 | Notebook burn: cell index 1 unless skip burn  | Agent   |

## Phase 1 — Spec on disk (brief before moves)

| #   | Step                                         | Deliverable                   |
|-----|----------------------------------------------|-------------------------------|
| 1.1 | Write sports/540_READ_ME_SIM.md              | Re-entry sim contract         |
| 1.2 | Pointer in re_entry/00_READ_ME_FIRST.md      | One paragraph                 |
| 1.3 | Update re_entry/02, 03: τ → ρ for assignment | Short patches                 |
| 1.4 | Update PARKED_FOR_LATER.md                   | 538D archived; active = 540_* |

Gate: Charles approves 540\_READ\_ME\_SIM.md before Phase 2.

## Phase 2 — Archive

| #   | Step                               | Deliverable                                                                                                                                                                            |
|-----|------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2.1 | Create sports/archive/ + README.md | Folder + manifest                                                                                                                                                                      |
| 2.2 | Move into archive                  | 538D_development.ipynb, 538_alex_tier1_model_and_fit.ipynb, 537_Sports_Simulation.ipynb, 535_sports_tier_1.ipynb, 538D_widget_render_test.ipynb, optional 535_*_outputs_backup_*.ipynb |
| 2.3 | Grep and fix broken doc pointers   | Docs we touch only                                                                                                                                                                     |
| 2.4 | Do not move                        | 530, tier1_*.py, hero_model_reset_bundle.py, sports_pipeline/                                                                                                                          |

## Phase 3 — COMPASS touch (light)

| #   | Step                                                 | Deliverable                         |
|-----|------------------------------------------------------|-------------------------------------|
| 3.1 | 3-Master_Plan/20260727_COMPASS_sim_reentry_status.md | ½-page status + claims guard        |
| 3.2 | Mirror plan if ~/.cursor/plans/ edited               | ./scripts/mirror_plan.sh (optional) |

No mandatory COMPASS agent session.

Phase 4 — SCOUT (reference only)

| #   | Step                                                      | Deliverable                             |
|-----|-----------------------------------------------------------|-----------------------------------------|
| 4.1 | No SCOUT session required this sprint                     | —                                       |
| 4.2 | Archived notebooks = SCOUT-era lab reference; do not edit | Note in <code>540_READ_ME_SIM.md</code> |
| 4.3 | Fresh SCOUT session only if <code>tier1_*</code> bug      | Uses 540 brief                          |

Phase 5 — Code (p + Pass B)

| #   | Step                                                    | Deliverable                                                                |
|-----|---------------------------------------------------------|----------------------------------------------------------------------------|
| 5.1 | p in <code>tier1_pool_assignment.py</code>              | <code>exp(-p · (A_i-T_j)²/(2σ²))</code> ; p=0 → uniform among open rosters |
| 5.2 | <code>USE_PREFERENTIAL_ATTACHMENT</code> bool in config | α=0 default                                                                |
| 5.3 | <code>sports/scripts/540_rho_ablation_bundle.py</code>  | Low p, high p, sort-and-chop; fix selection                                |
| 5.4 | Export <code>alex_rho_ablation_v0/</code>               | CSVs, PNG, summary, caption                                                |
| 5.5 | <code>sports/540_three_step_sim.ipynb</code>            | Thin Jupyter orchestration                                                 |
| 5.6 | Dry-run from repo root                                  | Green run                                                                  |

Pass A (λ knockout): already in `alex_side_by_side_v0/` — do not redo.

Pass B (p ablation): fix S\_i rule; vary assignment only.

Phase 6 — Reconcile Alex bundle

| #   | Step                                         | Deliverable              |
|-----|----------------------------------------------|--------------------------|
| 6.1 | Keep <code>alex_side_by_side_v0/</code>      | Pass A                   |
| 6.2 | <code>alex_rho_ablation_v0/README.txt</code> | Cross-link to λ knockout |
| 6.3 | Limitation sentence for p pass               | caption.txt              |

Phase 7 — Presentation to Alex (Charles leads)

| #   | Step                                    | Deliverable                          |
|-----|-----------------------------------------|--------------------------------------|
| 7.1 | <code>Model.pdf</code> + 2 min addendum | Three-step pipeline; Pass A + Pass B |
| 7.2 | Minimal table                           | λ knockout vs p ablation arms        |
| 7.3 | PDFs of updated docs                    | Charles runs convert scripts locally |
| 7.4 | Email / meeting                         | Figure paths + limitation sentences  |

Phase 8 — Close-out (when Charles asks)

| #   | Step                                                    |
|-----|---------------------------------------------------------|
| 8.1 | Git commit (docs + archive + scripts + tracked exports) |
| 8.2 | Park stretch in <code>PARKED_FOR_LATER.md</code>        |

Already done (skip)

- Empirical hero + LPM
- Pass A: λ knockout + `alex_side_by_side_v0/`
- Re-entry docs 01–03 + `Model.pdf`
- BINDING, unified S\_i, (B–D)=–L\_C

Execution order

1.1 → 1.4 → 2.1 → 2.4 → 3.1 → 5.1 → 5.6 → 6.1 → 6.3 → 7.x → 8.x (optional)

Assignment formulas (540 contract)

Three-step assignment (generative step 1):

- 1. Draw **A\_i** (abilities)
- 2. Draw **T\_j** (team target means)
- 3. Soft place:  $\pi_{ij} \propto \exp\left(-\rho\left(A_i - T_j\right)^2 / \left(2\sigma^2\right)\right) \times \text{optional } \left(n_j + k\right)^\alpha$

| $\rho$ | Meaning                                               |
|--------|-------------------------------------------------------|
| 0      | Uniform among teams with roster room (max mixing)     |
| 1      | Moderate ( $\equiv$ legacy $\tau=\sigma\approx0.65$ ) |
| High   | Sharper match to <b>T_j</b> (not sort-and-chop)       |

**Sort-and-chop**: separate benchmark ( `method="sort_chop"` ) — max assortativity in 537 sense; not  $\rho\rightarrow\infty$ .

**Selection (steps 2–3)**:  $S_i = A_i - \lambda \cdot L_C$ ; Pass B fixes  $\lambda$ , **L\_C**, K.

**Anti-patterns**

- Adding cells to archived notebooks
- Duplicating `tier1_*` logic inside 540
- Claiming  $\rho$  ablation reproduces hero bin-for-bin
- Renaming  $\rho$  back to  $\tau$  in user-facing prose

See also: `sports/540_READ_ME_SIM.md` , `3-Master_Plan/re_entry/00_READ_ME_FIRST.md`